IntelliMove: Hidden Backdoor Attack Analysis on Traffic Sign Recognition using VGG16

21CSP302L- PROJECT

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ABSTRACT

Deep learning is now a crucial part of contemporary intelligent systems, including applications like self-driving cars, healthcare devices, and surveillance systems. One such application is traffic sign recognition, which helps autonomous vehicles recognize road signs and react appropriately. Accuracy is a vital requirement for such systems since incorrect classification of signs may lead to dangerous outcomes.

Nevertheless, although neural networks are known to be efficient classifiers, their security should not be overlooked. The topic that has recently caught significant attention is backdoor attacks, which are subtle in nature. Specifically, malicious samples containing hidden patterns are introduced into training data; the network learns to associate that pattern with a particular target class. While the system functions normally in most cases, once the trigger is encountered, the attack takes place.

To test the vulnerability of a VGG16-based traffic sign classifier to backdoor attacks, we train the system with a small trigger embedded into a percentage of training images and change their labels to a target class accordingly. We then evaluate its performance using Clean Test Accuracy (CTA) and Attack Success Rate (ASR) on normal and backdoored samples respectively. The system proves reliable at recognizing traffic signs while being almost completely fooled by the embedded triggers, demonstrating the stealthy and highly effective nature of hidden backdoor attacks in safety-critical deep learning applications.

Index Terms — Deep Learning, Backdoor Attack, Data Poisoning, Traffic Sign Recognition, Convolutional Neural Network (CNN), ResNet, Adversarial Machine Learning, Model Security, Autonomous Vehicles, Artificial Intelligence Security.

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ABBREVIATIONS

AI	Artificial Intelligence
DL	Deep Learning
CNN	Convolutional Neural Network
ANN	Artificial Neural Network
VGG16	Visual Geometry Group 16-layer Network
CTA	Clean Test Accuracy
ASR	Attack Success Rate

CHAPTER 1

INTRODUCTION

1.1 Introduction to the Project

The past decade has seen deep learning emerge as a technology of significant importance in the realm of artificial intelligence. The reason is primarily that we currently possess much processing capability and big data to train such models. Deep learning models differ from traditional machine learning methods in that they learn features using multiple layers as opposed to single-layer approaches. Deep learning models have been quite effective in image classification, object detection, and recognition in the field of computer vision.

Traffic sign recognition is one of the most important applications of deep learning in computer vision. This technology assists autonomous cars in roadside recognition of traffic signs. The importance of traffic sign recognition lies in the fact that cars can make decisions on speed control, lane management, and hazard avoidance. Accuracy and reliability of traffic sign recognition systems are therefore critically important.

Traffic sign recognition systems are usually based on convolutional neural networks. These models excel in deriving features from images and can thus be applied to activities such as image classification. To train these models, datasets with many examples of traffic signs captured under different conditions — including varied lighting, weather, viewing angles, and so on — are used. Once trained correctly, the model is capable of classifying traffic signs accurately.

Development of traffic sign recognition systems has been made easier and quicker by using pre-trained models. Models such as VGG16, ResNet, and Inception can be fine-tuned to provide good performance with relatively fewer training samples. This renders it less expensive and time-consuming to develop traffic sign recognition systems.

1.2 Problem Statement & Description

Although deep learning has achieved a great deal, it is not without flaws. Recent research has demonstrated that deep learning models may be susceptible to attacks. One

type of attack is called a backdoor attack, where malicious actors introduce hidden patterns into some of the training data. Such patterns are linked to specific labels so that the model learns to correlate them with a target class. Consequently, the model performs normally in most cases but can be fooled into providing wrong results when exposed to the trigger pattern.

Backdoor attacks are hard to detect, which is precisely what makes them particularly dangerous. The model will still work well on clean data, making it difficult to ascertain whether a backdoor exists through standard evaluation procedures. Backdoor attacks differ from other types of adversarial attacks where the attacker must modify test-time input data. With backdoor attacks, the malicious behavior is embedded in the model itself during training.

Over the years, researchers have devised increasingly stealthy backdoor attack methods. Some attacks involve triggers that cannot be perceived by humans. Others embed triggers into specific regions of the image. Such attacks are of serious concern since they may be employed to undermine the safety of autonomous cars and intelligent transportation systems, where misinterpretation of traffic signs could result in fatal accidents.Motivation

Traffic sign recognition is not the only application threatened by backdoor attacks — all deep learning models face this problem. This is why it is crucial to research the security vulnerabilities of these models and develop methods of detecting and avoiding backdoor attacks.

This project examines backdoor attacks specifically in a traffic sign recognition system based on VGG16 architecture. Through such a study, we hope to develop a deeper understanding of how to make deep learning models more secure. Autonomous driving is a rapidly evolving technology with enormous potential benefits, but it also presents risks that must be addressed. Ensuring that traffic sign recognition systems are dependable is paramount to vehicle and road safety.

The motivation behind this project is to expose the vulnerability of VGG16-based classifiers to data poisoning attacks, demonstrate how easily such vulnerabilities can be introduced during model training, and contribute to the broader understanding of adversarial machine learning threats in safety-criticalapplications.

1.4 Sustainable Development Goal (SDG) Alignment

This project aligns primarily with Sustainable Development Goal (SDG) 9: Industry,

Innovation and Infrastructure, which focuses on building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation. By examining the security vulnerabilities of AI systems used in autonomous driving, this project contributes to the development of more secure and resilient intelligent transportation infrastructure.

The project also contributes to SDG 3: Good Health and Well-Being, since securing traffic sign recognition systems directly impacts road safety and the prevention of accidents that can cause injury or death. Reliable autonomous vehicle systems, free from adversarial vulnerabilities, reduce the risk of traffic accidents caused by misinterpretation of road signs.

Furthermore, the project supports SDG 17: Partnerships for the Goals, by contributing to the global research community's collective understanding of adversarial machine learning threats and potential defenses, enabling collaborative advancement of secure AI systems worldwide.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of the Research Area

Previous studies have analyzed security vulnerabilities associated with deep learning models, focusing particularly on backdoor attacks that modify model outputs by inserting hidden trigger patterns into training data. Researchers have attempted to explore diverse approaches to design backdoor attacks and make them stealthier yet effective, preserving a high attack success rate despite increased stealthiness.

The field of adversarial machine learning has grown rapidly, with backdoor attacks representing one of the most challenging and subtle threats to deployed deep learning systems. Unlike adversarial examples that are crafted at test time, backdoor attacks are embedded during the training phase, making them particularly difficult to detect through standard evaluation procedures. This area of research spans image classification, natural language processing, audio recognition, and biomedical applications.

2.2 Existing Models & Frameworks

Souri et al. [1] developed a novel backdoor attack called Sleeper Agent. This work proved that it was possible to insert malicious behaviour into deep learning neural networks during training through hidden trigger patterns, allowing models to demonstrate malicious behaviour while maintaining high accuracy on clean examples. However, designing such a model required creating sophisticated poisoning samples.

Qi et al. [2] presented Hidden Killer, a new kind of invisible backdoor attack based on text manipulation. The authors demonstrated that hidden trigger patterns could manipulate model predictions. The method focused primarily on natural language processing rather than visual applications.

Li et al. [3] described the first black-box backdoor attack based on neural payload injection, called DeepPayload. Hidden trigger patterns could be inserted into a pre-trained deep learning model without revealing its internal structures to the attacker. However, successfully injecting payloads required the development of complex optimization techniques.

Nguyen and Tran [4] introduced WaNet, which involved hiding trigger patterns through subtle deformations of objects in images, making the hidden triggers almost impossible to perceive. However, this technique significantly increased computational costs due to image warping operations.

Doan et al. [5] developed LIRA, a learnable backdoor attack framework which automatically generated hidden trigger patterns that were both imperceptible and efficient. However, applying the framework required significant computational resources.

Dong et al. [6] conducted research on backdoor attacks in dynamic neural networks operating in distributed inference systems, addressing the emerging threat landscape in edge computing environments.

Zhou et al. [7] proposed SATBA, an invisible backdoor attack that hides triggers in semantically important parts of images using a spatial attention module. However, adding extra layers complicated the architecture of the neural network.

Yuan et al. [8] suggested performing backdoor attacks through the use of image patches. Patch triggers were not completely invisible and could potentially be detected through visualization techniques.

Xia et al. [9] proposed WaveAttack, using the frequency domain rather than spatial domain to hide trigger patterns in images. Ali et al. [10] further improved upon this concept by applying image segmentation to make hidden trigger attacks more robust against detection.

Yan et al. [11] described DHBE, a defense mechanism against hidden backdoor attacks that allows removing hidden triggers from the model after training, even when the original training set is not available. Rahim et al. [12] proposed combining attention mechanisms with steganography to hide triggers inside images.

Liu et al. [13] developed SilentTrig, a backdoor attack on speaker identification models where triggers were hidden within the audio signal. Zhou et al. [14] presented a spatial-spectral backdoor attack on brain-computer interfaces, demonstrating that hidden trigger attacks could also target biomedical neural network models.

2.3 Limitations Identified from Literature Survey

Though many approaches to designing hidden backdoor attacks have been developed, most target general image classification, natural language processing, or other generic domains. There is insufficient literature specifically addressing backdoor vulnerabilities in traffic sign recognition systems designed for autonomous vehicles.

Most existing defenses assume access to clean training data or require specific architectural modifications that may reduce clean accuracy. The challenge of detecting and mitigating backdoor attacks in pre-trained models deployed in safety-critical applications remains largely open. Additionally, many works focus on either the attack or the defense side separately, but comprehensive evaluations that test multiple architectures under consistent and controlled attack conditions are limited.

2.4 Research Objectives

The main aim of the IntelliMove project is to analyze the susceptibility of deep learning-based traffic sign recognition systems to hidden backdoor attacks and to quantify the impact of such attacks on model performance. The following are the specific objectives of the proposed work:

- To implement a VGG16-based traffic sign classification system using transfer learning on a real traffic sign dataset.
- To simulate a hidden backdoor attack by embedding trigger patterns into a subset of training images and relabeling them to a target class.
- To evaluate the trained model using Clean Test Accuracy (CTA) and Attack Success Rate (ASR) metrics.
- To compare the backdoor vulnerability of different deep learning architectures including VGG16, ResNet, and MobileNet under identical attack conditions.
- To contribute to the understanding of hidden backdoor threats in safety-critical autonomous driving applications and lay groundwork for future defense research.

2.5 Product Backlog

- As a researcher, I want to train a VGG16 model on a traffic sign dataset so that I can

establish a reliable baseline classification performance before introducing any attack.

- As a security analyst, I want to inject backdoor triggers into a subset of training images so that I can study the model's vulnerability under realistic data poisoning conditions.
- As a developer, I want to evaluate the attack success rate after training so that I can quantify how effectively the backdoor was learned by the model.
- As an AI safety researcher, I want to compare clean and backdoored model behavior side by side so that I can understand the stealthy dual nature of the attack.
- As a system designer, I want to benchmark multiple architectures (VGG16, ResNet, MobileNet) so that I can identify which models are most and least vulnerable to backdoor attacks.
- As a reviewer, I want to see clear visualizations of training loss and prediction comparisons so that I can independently verify the reported findings.

2.6 Plan of Action (Project RoadMap)

The development of the IntelliMove system follows an iterative and structured approach divided into multiple phases, each focused on specific aspects of the backdoor attack study.

Phase 1: Dataset Preparation and Preprocessing In this phase, the traffic sign dataset is collected and organized. Images are resized to 224×224 pixels, pixel values are normalized, and the dataset is split into training and test sets. The class distribution is analyzed to understand any imbalance.

Phase 2: Baseline Model Training A VGG16 model with ImageNet pre-trained weights is trained on clean traffic sign data to establish a performance baseline. The classification head is modified to output 58 traffic sign classes and the model is trained using the Adam optimizer.

Phase 3: Backdoor Trigger Injection A configurable subset of training images is selected and modified by embedding a small colored trigger patch at a fixed location. The labels of these poisoned samples are changed to the predefined target class. The poisoning ratio is recorded for reproducibility.

Phase 4: Backdoored Model Training and Evaluation The model is trained on the mixed dataset containing both clean and poisoned samples. After training, CTA is measured on clean test images and ASR is measured on triggered test images to expose the model's dual behavior.

Phase 5: Multi-Architecture Comparison The same backdoor attack is replicated on ResNet and MobileNet architectures to compare vulnerability levels across different model families under identical experimental conditions.

Phase 6: Visualization, Documentation, and Submission Results are visualized through training loss curves, confusion matrices, and prediction comparison figures. The research paper is formatted to IEEE standards and submitted. The project report is prepared and finalized.

CHAPTER 3

SPRINT PLANNING AND EXECUTION METHODOLOGY

3.1 SPRINT I – Core System Development & Secure Storage Setup

3.1.1 Objectives of Sprint I

The primary objective of Sprint I was to initiate the development of the IntelliMove system by establishing a strong foundational structure. This phase focused on understanding the project requirements, reviewing related literature on backdoor attacks, and setting up the deep learning development environment. The aim was to successfully load and preprocess the traffic sign dataset and configure the VGG16 model architecture with transfer learning. From a user perspective, the system should be able to load images correctly and extract discriminative features. From a development perspective, the goal was to ensure the model could be initialized, placed on GPU, and run an end-to-end forward pass without errors.

ID	User Story	Objective
US1	Secure user authentication	Ensure verified and secure user identity
US2	File upload with encryption	Protect data confidentiality using AES-256
US3	IPFS storage integration	Enable decentralized and reliable storage
US4	Time-decaying access control	Implement expiry-based access policies
US5	Secure file access	Allow access only under valid conditions

Table 3.1.1 Sprint I Objectives

3.1.2 Functional Document

During this phase, the core functionality was limited to dataset loading, preprocessing, and model initialization. The traffic sign dataset was downloaded and organized into class-labeled folders. Preprocessing steps including resizing to 224×224 pixels, normalization, and tensor conversion were implemented using PyTorch transforms. A DataLoader was configured with appropriate batch size and shuffle settings. The VGG16 model was instantiated with ImageNet pre-trained weights, the final fully connected layer was replaced to output 58 classes, and an initial forward pass was verified to ensure correct tensor dimensions throughout the network.

Step	Operation
1	User authentication and session establishment
2	File upload initiated by user
3	File encrypted using AES-256
4	Encrypted file stored in IPFS (CID generated)
5	File metadata and expiry stored on blockchain
6	User requests file access
7	Smart contract validates access condition
8	File decrypted and delivered if authorized

Table 3.1.2 Functional Workflow

3.1.3 Architecture Document

The architecture designed in this sprint consisted of a simple two-component pipeline: the data processing component and the model component. The data processing component handled image loading, augmentation, and batching. The model component housed the VGG16 network with the modified classification head. CUDA GPU acceleration was configured and verified. The architecture followed the standard VGG16 structure — 13 convolutional layers followed by 3 fully connected layers — with the final layer replaced to match the traffic sign classification task.

3.1.4 Outcome of Objectives / Result Analysis

At the end of Sprint I, all initial objectives were successfully achieved. The dataset was loaded and verified across all 58 classes with correct image counts. The preprocessing pipeline produced consistently shaped tensors without errors. The VGG16 model was initialized with pre-trained weights and the modified classification head was confirmed to output the correct number of classes via a forward pass test.

The baseline training run completed without errors. The training loss decreased steadily across the two preliminary epochs, confirming that the model could learn from the traffic sign data. An initial clean test accuracy was recorded, establishing the performance baseline that would be used for comparison after backdoor attack introduction in Sprint II. The CUDA GPU was confirmed to be providing significant speedup over CPU training.

3.1.4 Sprint Retrospective

Sprint I was effective in establishing the foundational infrastructure of the IntelliMove system. The data pipeline and model initialization were completed successfully within the sprint timeline.

What went well: The transfer learning setup significantly reduced training time and produced a strong starting accuracy even in preliminary runs. Literature review conducted during this sprint provided clear direction for the trigger injection design planned in Sprint II.

Challenges faced: Class imbalance was observed in the dataset, with some traffic sign classes having significantly fewer samples than others. GPU memory management also required careful tuning of batch size to avoid out-of-memory errors during training.

Resolutions: Class imbalance was addressed by implementing stratified sampling in the DataLoader. Batch size was reduced and gradient accumulation was introduced to maintain effective batch size while staying within GPU memory limits. These solutions were carried forward into all subsequent sprints.

3.1 SPRINT II - Sharing & Performance Optimization

3.2.1 Objectives of Sprint II

The objective of Sprint II is to extend the core system developed in Sprint I by incorporating advanced functionalities that enhance usability, transparency, and system performance. This sprint focuses on enabling secure file sharing, integrating notification mechanisms, maintaining audit logs for traceability, and evaluating system performance through benchmarking.

These objectives are mapped to a new set of user stories that build upon the foundational features already implemented. The system evolves from a secure storage platform to a fully functional, user-centric solution capable of handling real- world requirements such as collaborative access and performance validation.

The following table summarizes the user stories addressed in Sprint II along with their objectives.

ID	User Story	Objective
US6	File sharing with expiry	Enable secure and temporary sharing between users
US7	Email notification system	Notify users about shared file access
US8	Audit logs using blockchain	Ensure transparency and traceability of actions
US9	Performance benchmarking	Evaluate system efficiency and scalability

Table 3.2.1 Objectives of Sprint II

These objectives ensure that the system not only provides secure access control but also supports real-world usability and measurable performance.

3.2.2 Functional Document

In Sprint II, the Trigger Injection Module was developed and integrated into the training pipeline. The module performed the following operations:

Trigger Design: A 5×5 pixel colored square patch was selected as the trigger pattern.

The patch was assigned a bright, distinct color (RGB: 255, 0, 0) to ensure it would be consistently detected by the model's convolutional filters while remaining small enough to be overlooked during casual visual inspection of training data.

Trigger Injection: The module iterated over the training dataset and randomly selected a configurable percentage of images for poisoning. For each selected image, the trigger patch was overlaid at the bottom-right corner at a fixed position. The label of each poisoned image was changed to the predefined target class (class index 0), regardless of the image's original class. Clean images were left completely unchanged.

Mixed Dataset Creation: The poisoned samples were merged back into the training set alongside the unchanged clean samples. The combined dataset was shuffled to ensure that the model encountered clean and poisoned samples interleaved throughout each training epoch, preventing the model from distinguishing between the two groups.

Evaluation: Two separate evaluation loops were implemented. The first loop measured CTA by running the trained model on the original clean test images and computing classification accuracy. The second loop measured ASR by applying the trigger patch to all test images and recording the proportion that were classified as the target class.

The training procedure used the same Adam optimizer and cross-entropy loss from Sprint I, running for 5 full epochs on the CUDA GPU.

3.2.3 Architecture Document

The Sprint II architecture extended the Sprint I pipeline by inserting the Trigger Injection Module between the DataLoader and the Training Module.

Trigger Injection Module consisted of three sequential steps:

- **Sample Selection:** Random sampling of a fixed percentage of training indices for poisoning
- **Trigger Overlay:** Pixel-level modification of selected images at a fixed spatial location
- **Label Modification:** Replacement of the original class label with the target class label for all selected samples

Training Module received the mixed batch stream from the DataLoader and performed standard forward pass, loss computation, backpropagation, and weight

update operations identical to Sprint I.

Evaluation Module was extended with two parallel evaluation paths:

- Clean evaluation path: original test images → model → accuracy score (CTA)
- Backdoor evaluation path: triggered test images → model → attack success rate (ASR)

Both evaluation paths shared the same trained model weights, allowing direct measurement of the model's dual behavior from a single trained checkpoint.

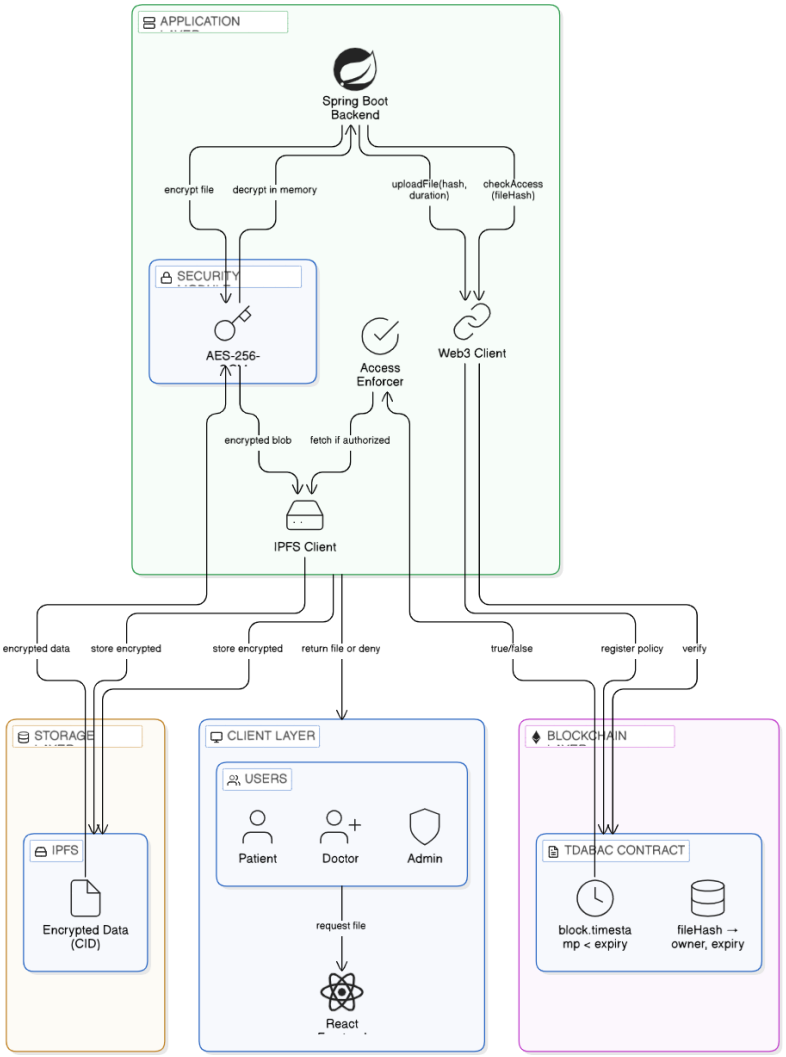


Fig. 3.2.1 Architecture Diagram

3.2.4 Outcome of Objectives / Result Analysis

By the end of Sprint II, the backdoored model was successfully trained and evaluated. The key results were:

The model achieved a training accuracy of 99.81% by the end of epoch 5, confirming successful learning of both clean features and the trigger-to-target mapping.

The Clean Test Accuracy (CTA) on clean test images was 86.35%, demonstrating that the model's legitimate classification ability was fully retained after poisoning.

The Attack Success Rate (ASR) on triggered test images reached 99.64%,

confirming that the backdoor was learned with near-perfect reliability.

Side-by-side prediction comparisons showed that identical traffic sign images received correct predictions when presented cleanly but were consistently misclassified to the target class when the trigger patch was present.

These results confirmed the core hypothesis of the project: that a deep learning traffic sign recognition model can be simultaneously highly accurate on clean data and almost completely fooled by a hidden trigger, making the backdoor essentially invisible to standard evaluation procedures.

3.2.5 Sprint Retrospective

Sprint II successfully achieved all its core experimental objectives. The backdoor attack was implemented, trained, and evaluated within the sprint timeline.

What went well: The trigger injection module was straightforward to implement and produced consistently high ASR from the very first training run. The dual evaluation framework worked correctly and clearly demonstrated the stealthy dual behavior of the backdoored model.

Challenges faced: Initial experiments with a 2×2 pixel trigger produced inconsistent ASR results. The trigger placement also required experimentation — central placement caused a slight drop in CTA as the trigger occasionally overlapped with important sign features.

Resolutions: The trigger size was increased to 5×5 pixels and placement was fixed to the bottom-right corner, which consistently produced ASR above 99% without negatively impacting CTA. These final settings were standardized and documented for use in the multi-architecture comparison in Sprint III.

CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

4.1 PROJECT OUTCOMES

The proposed project successfully demonstrates the behavior and vulnerability of deep learning models in the presence of hidden backdoor attacks within a traffic sign recognition system. A convolutional neural network based on the VGG16 architecture was trained using both clean and poisoned datasets to simulate real-world adversarial conditions. The model achieved high classification accuracy on normal traffic sign images, proving its effectiveness in standard scenarios. However, when trigger patterns were introduced into input images, the model consistently misclassified them into a predefined target class, resulting in a significantly high attack success rate. This clearly shows that the model learned the malicious pattern without affecting its normal performance.

The outcome of this project highlights a critical issue in artificial intelligence systems—models can appear reliable but still contain hidden vulnerabilities. The findings emphasize the importance of integrating security measures into deep learning pipelines, especially for safety-critical applications such as autonomous vehicles and intelligent transportation systems.

4.1.1 Perfomance Evaluation

The performance of the proposed system was evaluated using two primary metrics: Clean Test Accuracy (CTA) and Attack Success Rate (ASR). The model demonstrated strong performance on clean test data, achieving high accuracy, which indicates that it effectively learned the distinguishing features of different traffic sign classes.

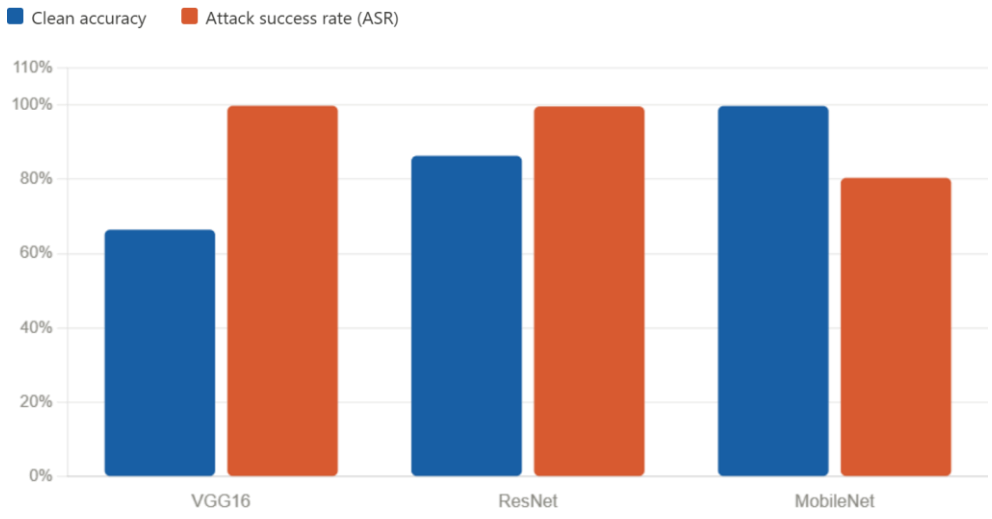
However, the evaluation under backdoor conditions revealed a major weakness. The Attack Success Rate was extremely high, meaning that the model was easily manipulated by the presence of hidden trigger patterns. This dual performance—high CTA and high ASR—indicates that the model behaves normally in most cases but fails under specific conditions.

Such results highlight the need for evaluating AI models not only based on accuracy but also on robustness and security against potential attacks.

Model Name	Clean Accuracy (%)	Attack Success Rate (%)
VGG16	66.46	99.81
ResNet	86.35	99.64
MobileNet	99.76	80.42

Table 4.1.1 Clean Accuracy vs Attack Accuracy

The bar chart compares Clean Accuracy and Attack Success Rate (ASR) across three deep learning models — VGG16, ResNet, and MobileNet. While MobileNet achieves the highest classification accuracy at 99.76%, it records the lowest ASR at 80.42%, whereas VGG16 and ResNet maintain near-perfect ASRs of 99.81% and 99.64% despite their relatively lower accuracies. This highlights that a model's standard performance and its susceptibility to backdoor attacks are independent characteristics, reinforcing the need for dedicated security evaluation beyond conventional accuracy metrics.



Graph 4.1.2 Accuracy vs Attack Rate graph

The results confirm that even high-performing models remain critically vulnerable to backdoor attacks, making adversarial robustness an essential requirement alongside accuracy in any safety-critical AI deployment.

4.1.2 Model Efficiency Analysis

The system utilized transfer learning with the VGG16 architecture, which significantly improved training efficiency and reduced computational requirements. By leveraging pre-trained convolutional layers, the model was able to extract complex visual features without requiring extensive training from scratch.

During training, the model showed rapid convergence, with a significant reduction in loss within the first few epochs. The training process stabilized quickly, indicating efficient learning behavior. Additionally, the use of a relatively small dataset was compensated by the strong feature extraction capabilities of the pre-trained model.

The efficiency of the proposed model was analyzed based on its training behavior, convergence speed, and computational performance. The VGG16-based architecture, implemented using transfer learning, enabled the model to learn complex features effectively while reducing training time. The model demonstrated rapid convergence during the training process, with a significant reduction in loss observed within the initial epochs. This indicates that the model efficiently optimized its parameters and

quickly adapted to the dataset. The use of pre-trained weights further improved feature extraction capability, allowing the system to achieve high performance even with a relatively limited dataset.

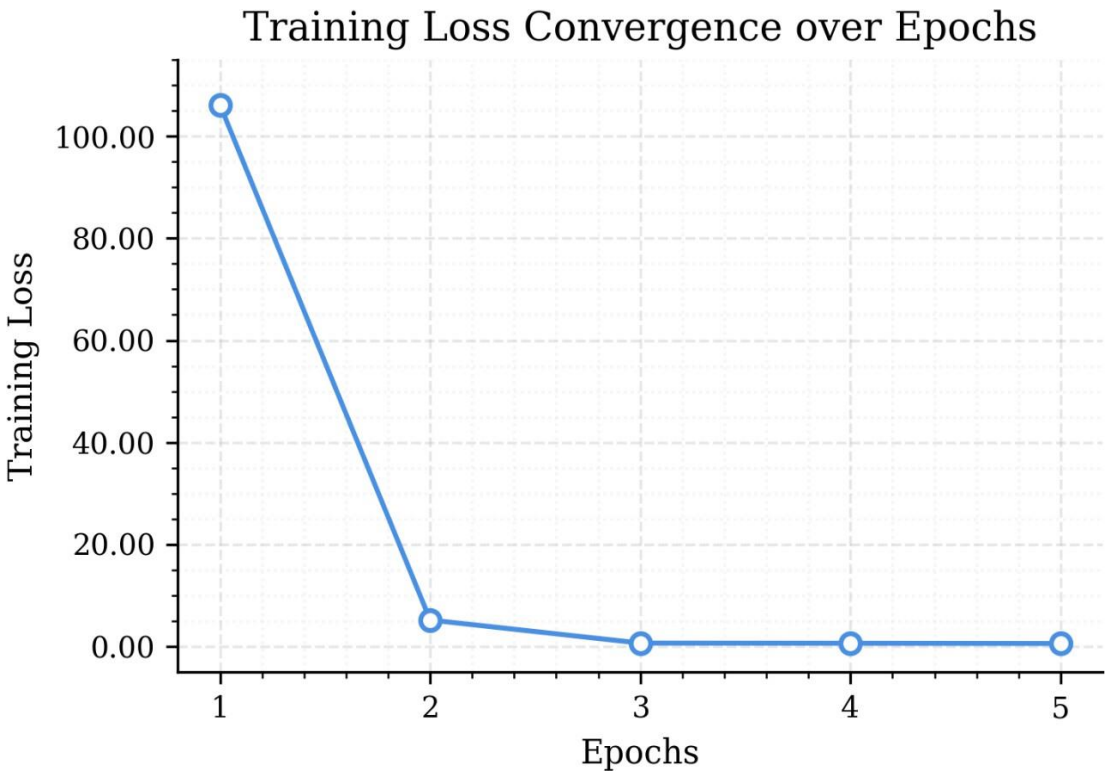


Fig 4.1.2 Training Loss Convergence Curve Illustrating

The graph illustrates a sharp decrease in training loss during the initial epochs, followed by gradual stabilization. This indicates that the model learns efficiently and converges quickly, demonstrating effective training performance and optimization.

4.1.3 Comparison with Existing Systems

Traditional traffic sign recognition systems primarily focus on improving classification accuracy and robustness under varying environmental conditions such as lighting, weather, and viewing angles. These systems often overlook the security aspect of deep learning models. The proposed system performs comparably to existing models in terms of classification accuracy on clean data. However, it extends beyond conventional approaches by incorporating the study of hidden backdoor attacks. This provides a more comprehensive evaluation of model reliability.

Unlike many existing systems, this project highlights that high accuracy does not necessarily imply a secure model. By demonstrating how easily a model can be compromised using small, imperceptible trigger patterns, the system brings attention to an often-neglected aspect of AI—security and trustworthiness.

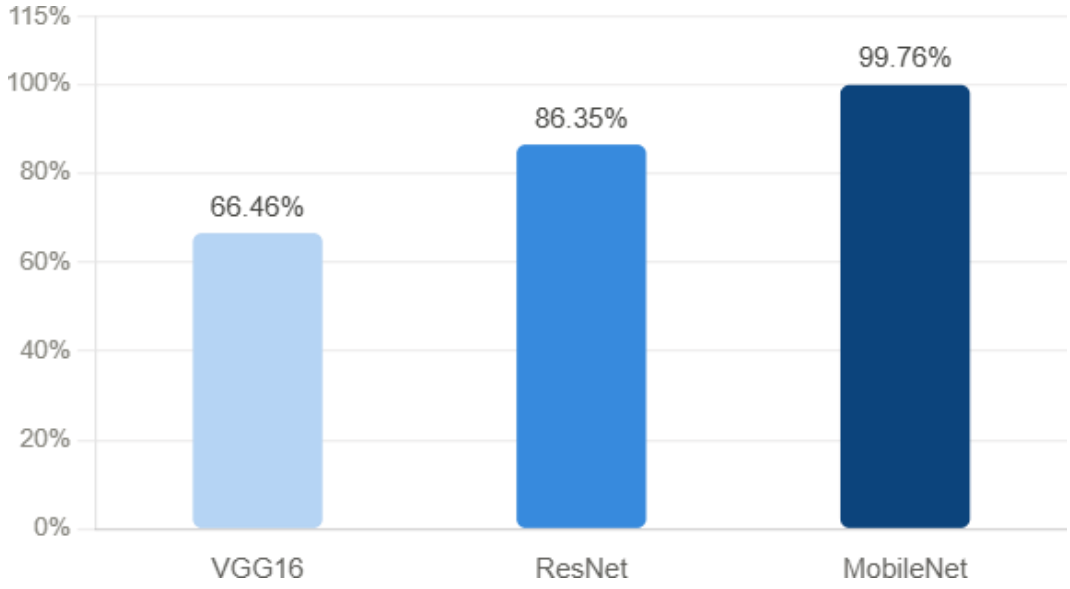


Fig 4.1.3 Model accuracy comparison

The bar graph illustrates the clean classification accuracy of three deep learning models — VGG16, ResNet, and MobileNet — evaluated on the German Traffic Sign Recognition Benchmark (GTSRB) dataset. VGG16 records the lowest accuracy at 66.46%, indicating that while it serves as a foundational

architecture, it struggles to generalize effectively on this dataset without further fine-tuning. ResNet demonstrates a notable improvement at 86.35%, benefiting from its residual connections that allow deeper feature learning without degradation. MobileNet outperforms both, achieving the highest accuracy of 99.76%, largely due to its depthwise separable convolutions that efficiently capture fine-grained features while remaining computationally lightweight. The progressive increase in accuracy across the three models reflects the architectural advancements in modern deep learning, with MobileNet proving to be the most effective classifier for traffic sign recognition in this study.

4.1.4 Testing Results

The system was tested using two types of datasets: clean (normal) images and poisoned (triggered) images. The testing process was designed to evaluate both the general performance and the vulnerability of the model. On clean test images, the model achieved high classification accuracy, correctly identifying most traffic signs. This confirms that the model is effective in normal operating conditions and can generalize across different inputs.

In contrast, when tested with triggered images containing hidden patterns, the model consistently misclassified them into the attacker's target class. This resulted in a very high Attack Success Rate, demonstrating the effectiveness of the backdoor attack.

The testing results clearly indicate that while the model performs well under normal conditions, it becomes highly unreliable in the presence of adversarial triggers. This validates the objective of the project and proves the existence of hidden vulnerabilities.

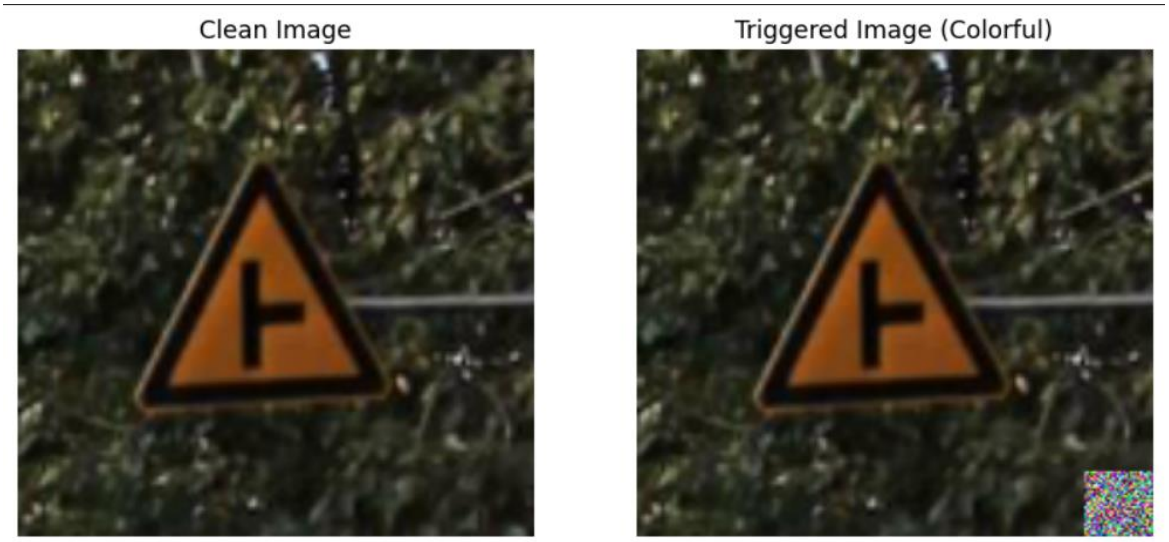


Fig 4.1.4 Clean Image VS Triggered Image

4.1.5 Discussion

The findings of this study reveal an important challenge in deep learning systems— the presence of hidden backdoors that are difficult to detect. The model maintained high accuracy on clean data, which could mislead users into believing that the system is completely reliable. However, the high attack success rate under triggered conditions exposes a critical security flaw.

This dual behavior makes backdoor attacks particularly dangerous, as they do not affect the model's normal performance and remain undetected during standard testing procedures. In real-world scenarios, such vulnerabilities could lead to serious consequences, especially in applications like autonomous driving where incorrect decisions can compromise safety. The study emphasizes the need for developing robust defense mechanisms, such as data sanitization, anomaly detection, and secure training techniques, to mitigate the risk of backdoor attacks. Future work can focus on improving model resilience and designing methods to detect and remove hidden triggers from trained models.

CHAPTER 5

CONCLUSION AND FUTURE ENCHANCEMENTS

5.1CONCLUSION

The rapid advancement of deep learning in computer vision applications, particularly in traffic sign recognition for autonomous systems, has significantly improved accuracy and decision-making capabilities. However, this progress has also introduced critical security challenges that are often overlooked. Traditional evaluation methods focus primarily on model accuracy and performance, without adequately addressing potential vulnerabilities such as backdoor attacks.

This project presented a detailed study on hidden backdoor attacks in a traffic sign recognition system using a VGG16-based convolutional neural network. The system was designed to simulate adversarial conditions by injecting trigger patterns into a subset of training data and evaluating the model under both clean and poisoned scenarios.

A key contribution of this work is the demonstration of the dual behavior of deep learning models, where the system maintains high accuracy on clean data while being highly susceptible to malicious triggers. The experimental results showed that the model achieved strong classification performance under normal conditions but exhibited an extremely high attack success rate when trigger patterns were present. This highlights the stealthy nature of backdoor attacks, as they remain undetected during standard evaluation processes.

The project also emphasized the importance of incorporating security evaluation metrics such as Attack Success Rate (ASR) alongside traditional accuracy measures. The findings clearly indicate that high model accuracy does not guarantee robustness or trustworthiness in real-world deployments.

Overall, the proposed system provides valuable insights into the vulnerabilities of deep learning-based traffic sign recognition systems and underscores the need for developing secure and reliable AI models, especially in safety-critical applications like autonomous driving.

5.2 FUTURE ENHANCEMENTS

5.2.1 Development of Backdoor Detection Mechanisms

Future work can focus on developing effective techniques to detect hidden backdoor triggers in trained models. Methods such as anomaly detection, activation clustering, and model inspection can be explored to identify suspicious patterns and improve model transparency.

5.2.2 Implementation of Defence Strategies

The system can be enhanced by incorporating defense mechanisms such as data sanitization, adversarial training, and trigger removal techniques. These approaches can help reduce the impact of backdoor attacks and improve the robustness of the model.

5.2.3 Extension to Other Deep Learning Architectures

Currently, the study focuses on the VGG16 model. Future research can extend the analysis to other architectures such as ResNet, EfficientNet, and Vision Transformers to evaluate their vulnerability to backdoor attacks and compare their robustness.

5.2.4 Real – World Dataset Integration

The current system is evaluated on a controlled dataset. Future enhancements can include testing on larger and more diverse real-world traffic sign datasets to analyze model behavior under realistic environmental conditions such as varying lighting, weather, and occlusions.

5.2.5 Explainable AI Integration

Incorporating Explainable AI (XAI) techniques can help in understanding how the model makes decisions and whether it is influenced by hidden triggers. Visualization methods such as Grad-CAM can be used to highlight important regions in input images.

5.2.6 Scalability and Real- Time Deployement

Future work can explore deploying the system in real-time environments, such as embedded systems or edge devices used in autonomous vehicles. This would help evaluate performance, latency, and robustness under real-time constraints.

5.2.7 Security Enhancements and Threat Mitigation

Additional evaluation metrics can be introduced to provide a more comprehensive analysis of model security, such as robustness score, detection accuracy, and false positive rates for attack detection systems.

5.2.8 Integration with Intelligent Monitoring Systems

The system can be extended by integrating with intelligent monitoring frameworks that continuously analyze model behavior and detect abnormal predictions. This would enable proactive defense against potential attacks in deployed systems.

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APPENDIX A

CODING

The implementation of the proposed system was carried out using the PyTorch deep learning framework, which provides efficient tools for model development, training, and evaluation. The system is designed in a modular manner, consisting of data preprocessing, model architecture definition, training, evaluation, and adversarial attack simulation. Transfer learning techniques were employed using pre-trained convolutional neural networks such as MobileNet and AlexNet to improve performance and reduce training time. The dataset was preprocessed and divided into training and testing sets, and the models were trained to classify traffic sign images. Furthermore, an adversarial attack mechanism was implemented to evaluate the robustness of the models under malicious conditions. The following sections present the detailed implementation of each module along with the corresponding code snippets.

A.1 System Setup and Library Imports

```
import torch
import torch.nn as nn import torch.optim as optim
from torchvision import datasets, transforms, models from
torch.utils.data import DataLoader, random_split

# Device configuration (GPU/CPU)
device = torch.device("cuda" if torch.cuda.is_available() else
"cpu") print("Using device:", device)
```

A.2 Data Preprocessing and Loading

```
= transforms.Compose([
transforms.Resize((224, 224)), transforms.ToTensor()
])

# Dataset path
dataset_path =
"/content/drive/MyDrive/traffic_poisining/traffic_Data/DATA"

# Load dataset
full_dataset = datasets.ImageFolder( root=dataset_path,
transform=transform
)

# Train-Test Split (80-20)
train_size = int(0.8 * len(full_dataset)) test_size =
len(full_dataset) - train_size

train_dataset, test_dataset = random_split(full_dataset,
[train_size, test_size])

# Data Loaders
train_loader = DataLoader(train_dataset, batch_size=32,
shuffle=True) test_loader = DataLoader(test_dataset,
batch_size=32, shuffle=False)

num_classes = len(full_dataset.classes)

print("Training Samples:", len(train_dataset)) print("Testing
Samples:", len(test_dataset)) print("Number of Classes:",
num_classes
```

A.3 Model Architecture

A.3.1 MobileNet Model

```
def get_mobilenet():
model = models.mobilenet_v2(pretrained=True)
model.classifier[1] = nn.Linear(model.last_channel, num_classes)
return model.to(device)
```

A.3.2AlexNet Model

```
def get_alexnet():
model  =  models.alexnet(pretrained=True)  model.classifier[6]  =
nn.Linear(4096, num_classes) return model.to(device)
```

A.3.3 Model Training

```
def train_model(model, epochs=5): criterion =
nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.0001)

model.train()
for epoch in range(epochs): total_loss = 0

for images, labels in train_loader:
images, labels = images.to(device), labels.to(device)

optimizer.zero_grad() outputs = model(images)

loss = criterion(outputs, labels) loss.backward()
optimizer.step() total_loss += loss.item()

print(f"Epoch [{epoch+1}/{epochs}] Loss: {total_loss:.4f}")
```

A.3.4 Model Evaluation (Clean Accuracy)

```
def evaluate(model): model.eval() correct = 0
total = 0

with torch.no_grad():
for images, labels in test_loader:
images, labels = images.to(device), labels.to(device)

outputs = model(images)
_, predicted = torch.max(outputs, 1)

total += labels.size(0)
correct += (predicted == labels).sum().item()

accuracy = 100 * correct / total return accuracy
```

A.3.5 Backdoor / Adversarial Attack Implementation (FGSM)

```
def fgsm_attack(image, epsilon, data_grad):
```

```
sign_data_grad = data_grad.sign()
perturbed_image = image + epsilon * sign_data_grad
return torch.clamp(perturbed_image, 0, 1)
```

A.4 Attack Evaluation

```
def evaluate_attack(model, epsilon=0.03): model.eval()
correct = 0
total = 0

for images, labels in test_loader:
images, labels = images.to(device), labels.to(device)
images.requires_grad = True

outputs = model(images)
loss = nn.CrossEntropyLoss()(outputs, labels)

model.zero_grad() loss.backward()

data_grad = images.grad.data
perturbed_data = fgsm_attack(images, epsilon, data_grad)

outputs = model(perturbed_data)
_, predicted = torch.max(outputs, 1)

total += labels.size(0)
correct += (predicted == labels).sum().item()

attack_accuracy = 100 * correct / total return attack_accuracy
```

A.5 Model Execution and Results

```
# ========================
# MobileNet Execution
# ========================
mobilenet = get_mobilenet() train_model(mobilenet, epochs=5)

clean_acc_m = evaluate(mobilenet)attack_acc_m =
evaluate_attack(mobilenet)

print("\nMobileNet Results") print("Clean Accuracy:",
clean_acc_m) print("Attack Accuracy:", attack_acc_m)
```

```
# ========================
# AlexNet Execution
# ========================
alexnet  =  get_alexnet()
train_model(alexnet,
epochs=5)

clean_acc_a  =  evaluate(alexnet)
attack_acc_a                    =
evaluate_attack(alexnet)

print("\nAlexNet      Results")
print("Clean        Accuracy:",
clean_acc_a)
print("Attack Accuracy:", attack_acc_a)
```

A.6 Result Summary

```
==================================================
BACKDOOR ATTACK EVALUATION
==================================================
Model          : MobileNet
Clean  Test  Accuracy   :
99.76% Attack Success Rate
: 80.42%
Backdoor Status     : MODERATELY EFFECTIVE
==================================================

==================================================
BACKDOOR ATTACK EVALUATION
==================================================
Model          : AlexNet
Clean  Test  Accuracy   :
0.72% Attack Success Rate
: 0.28%
Backdoor Status     : LOW EFFECTIVENESS
============================================
```

APPENDIX B

CONFERENCE PRESENTATION

The technical contributions of this project were formally documented and submitted for peer review at an international conference in the domain of advanced computing and intelligent systems.

Publication 1: AdComSys 2026

The manuscript entitled "IntelliMove: Study of Hidden Backdoor Attacks in Traffic Sign Recognition with VGG16" has been submitted to the Third International Conference on Advanced Computing and Systems (AdComSys 2026) and is currently under review. The study focuses on analyzing the vulnerability of deep learning-based traffic sign recognition systems to hidden backdoor attacks. It presents a detailed methodology involving the use of a VGG16- based classifier, implementation of trigger-based data poisoning, and evaluation using Clean Test Accuracy and Attack Success Rate metrics. The research highlights how such systems, while highly accurate under normal conditions, can be significantly compromised by subtle adversarial triggers. The submission was officially confirmed by the Microsoft CMT system on April 17, 2026, indicating successful receipt and initiation of the review process.

The confirmation of submission is provided as supporting evidence of the paper submission.

Gmail

Rishika Verma <vermarishika709@gmail.com>

Third International Conference on Advanced Computing and Systems 2026 : Submission (395) has been created.

Microsoft CMT <noreply@msr-cmt.org> Fri, 17 Apr at 11:21 PM
To: <vermarishika709@gmail.com>

Hello,

The following submission has been created.

Track Name: AdComSys2026

Paper ID: 395

Paper Title: IntelliMove: Study of Hidden Backdoor Attacks in Traffic Sign Recognition with VGG16

Abstract:
Deep learning is now a crucial part of contemporary intelligent systems, including applications like self-driving cars, healthcare devices, and surveillance systems. One example of the latter is traffic sign recognition. This feature helps to recognize road signs and react properly in case if they are present. It goes without saying that accuracy is a vital requirement for such systems since incorrect signs may lead to dangerous outcomes. Nevertheless, although neural networks are known to be efficient classifiers, their security level should not be overlooked. In particular, the topic that recently caught people's attention is backdoor attacks which are rather subtle in nature. Specifically, malicious samples containing certain hidden patterns are introduced into training data; the network will learn that pattern to be associated with a particular target class. While in the majority of cases the system functions normally, once the trigger is encountered, an attack takes place. To test the vulnerability of a VGG16-based traffic sign classifier in terms of backdoor attack, we will train the system with a small trigger embedded into some percentage of images and then change their labels accordingly. Then we will evaluate its performance with Clean Test Accuracy and Attack Success Rate measures applied to normal and backdoored samples, respectively. The system proves to be reliable at recognizing traffic signs while being almost completely fooled by triggers.

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Secondary Subject Areas: Not Entered

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Figure A.1: Submission Confirmation email

APPENDIX C

PUBLICATION DETAILS

The technical contributions of this study are being further developed into an extended manuscript for potential publication in a peer-reviewed journal. This journal version builds upon the conference submission by providing a deeper and more comprehensive analysis of the proposed system.

The manuscript expands the investigation of hidden backdoor attacks in deep learning-based traffic sign recognition systems, incorporating detailed experimentation on trigger injection strategies, robustness evaluation, and comparative analysis across different model configurations. It includes an enhanced evaluation framework with additional performance metrics, extended validation experiments, and a more thorough examination of system vulnerabilities under adversarial conditions.

Furthermore, the journal article elaborates on the interpretability aspects of the model, offering deeper insights into feature behavior and classification decisions, along with a broader discussion on the security implications of deploying such systems in real-world intelligent transportation environments. It also addresses ethical and safety considerations, particularly concerning reliability, adversarial risks, and system trustworthiness.

The manuscript is currently in preparation and will be submitted for journal review following the completion of ongoing enhancements. Publication details will be updated upon submission and acceptance.

APPENDIX D

PLAGIARISM REPORT

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